Assignment No 8

Title :

To identify trusted Wi-Fi networks and detect fake access points for security

Analyze Wi-Fi network information to identify trusted networks and detect possible fake access points using SSID, BSSID, and security parameters.

1. Implementation

To perform the problem statement, Wi-Fi information available on a Windows laptop was inspected using the built-in netsh WLAN commands. The analysis focused on SSID, BSSID, authentication, encryption, signal strength, radio type, band, and channel.

- The connected Wi-Fi network was first identified as the known/trusted network for the observation.
- The nearby wireless networks were then scanned using BSSID-level information.
- The observed parameters were compared to distinguish the known network from an unknown network.
- An unknown network was treated as potentially suspicious only when its characteristics required further verification; the scan alone does not prove that an access point is fake.

2. Important Commands

2.1 Display Connected Wi-Fi Interface

```
netsh wlan show interfaces
```

This command displays the currently connected wireless interface and the connection parameters. In the performed test, the interface was connected to the SSID "motorola edge 60 fusion".

2.2 Scan Nearby Wi-Fi Networks with BSSID Information

```
netsh wlan show networks mode=bssid
```

This command lists currently visible wireless networks and provides BSSID-specific details such as signal, radio type, band, channel, authentication, and encryption.

3. Performance / Practical Execution

The following screenshots are the actual outputs obtained during the practical execution on the laptop.

3.1 Connected Network Details

```
Microsoft Windows [Version 10.0.26200.9168]
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C:\Users\hites>netsh wlan show interfaces

There is 1 interface on the system:

    Name                   : Wi-Fi
    Description            : Intel(R) Wi-Fi 6E AX211 160MHz
    GUID                   : 1cca7b86-e4f1-4b69-9f0c-bfd11908a892
    Physical address       : e0:2e:0b:f7:17:eb
    Interface type         : Primary
    State                  : connected
    SSID                   : motorola edge 60 fusion
    AP BSSID               : 76:18:9f:66:94:30
    Band                   : 5 GHz
    Channel                : 36
    Connected Akm-cipher   : [ akm = 00-0f-ac:08, cipher =  00-0f-ac:04 ]
    Network type           : Infrastructure
    Radio type             : 802.11ax
    Authentication         : WPA3-Personal
    Cipher                 : CCMP
    Connection mode        : Auto Connect
    Receive rate (Mbps)    : 432
    Transmit rate (Mbps)   : 480
    Signal                 : 80%
    Rssi                   : -64
    Profile                : motorola edge 60 fusion
    QoS MSCS Configured         : 0
    QoS Map Configured          : 0
    QoS Map Allowed by Policy   : 0
```

Observed Details

Parameter	Observed Value
Interface	Wi-Fi
Wi-Fi Adapter	Intel(R) Wi-Fi 6E AX211 160MHz
State	connected
SSID	motorola edge 60 fusion
AP BSSID	76:18:9f:66:94:30
Band	5 GHz
Channel	36
Radio Type	802.11ax
Authentication	WPA3-Personal
Cipher	CCMP
Connection Mode	Auto Connect
Signal	80%
RSSI	-64
Receive / Transmit Rate	432 Mbps / 480 Mbps

3.2 Nearby Network Scan

```
C:\Users\hites>netsh wlan show networks mode=bssid

Interface name : Wi-Fi
There are 2 networks currently visible.

SSID 1 : motorola edge 60 fusion
    Network type            : Infrastructure
    Authentication          : WPA3-Personal
    Encryption              : CCMP
    BSSID 1                 : 76:18:9f:66:94:30
         Signal             : 83%
         Radio type         : 802.11ax
         Band               : 5 GHz
         Channel            : 36
         Details            :  (H2E Not supported)
         QoS MSCS Supported    : 0
         QoS Map Supported     : 1
         Basic rates (Mbps) : 6 12 24
         Other rates (Mbps) : 9 18 36 48 54

SSID 2 : _Sumeetspeaks_
    Network type            : Infrastructure
    Authentication          : WPA2-Personal
    Encryption              : CCMP
    BSSID 1                 : 02:c3:42:9e:a2:7e
         Signal             : 86%
         Radio type         : 802.11ac
         Band               : 2.4 GHz
         Channel            : 1
         QoS MSCS Supported    : 0
         QoS Map Supported     : 0
         Basic rates (Mbps) : 1 2 5.5 11
         Other rates (Mbps) : 6 9 12 18 24 36 48 54
```

4. Analysis

The scan detected two currently visible networks. The connected network, "motorola edge 60 fusion", was observed with BSSID 76:18:9f:66:94:30, WPA3-Personal authentication, CCMP encryption, 5 GHz band, channel 36, and signal strength of 83% in the BSSID scan. These values match the connected-network information shown in the first screenshot.

The second visible network, "_Sumeetspeaks_", had BSSID 02:c3:42:9e:a2:7e, WPA2-Personal authentication, CCMP encryption, 2.4 GHz band, channel 1, and signal strength of 86%. Its SSID is different from the connected network, so the provided scan does not show a same-SSID duplicate for the trusted network.

Based only on the captured output, "motorola edge 60 fusion" can be treated as the known network used for this practical because it is the network to which the laptop was connected. The second network is an unknown nearby network, but the available evidence is not sufficient to label it as a fake access point. For reliable identification, the observed BSSID and security configuration should be compared with the known details of the legitimate network.

5. Result

- The laptop successfully displayed the connected Wi-Fi interface and its security parameters.
- The connected network was identified using its SSID and BSSID.
- Two nearby Wi-Fi networks were detected using BSSID-level scanning.
- The scan provided the required SSID, BSSID, authentication, encryption, signal, band, and channel information.
- No same-SSID duplicate access point was visible in the captured scan.
- The captured evidence is sufficient to demonstrate Wi-Fi network identification, but it is not by itself sufficient to prove that an unknown access point is fake.

6. Conclusion

The practical successfully demonstrated how Wi-Fi network information can be collected and analyzed on a Windows laptop for a mobile security assignment. The SSID identifies the visible network name, while the BSSID identifies the particular access point. Authentication, encryption, band, channel, and signal information provide additional characteristics for comparison. In this execution, the connected network "motorola edge 60 fusion" was observed with WPA3-Personal authentication and CCMP encryption, while a second network named "_Sumeetspeaks_" was also visible. The results demonstrate that trusted-network identification should use multiple parameters rather than relying on the SSID alone.